

NAME:SEJAL KALBHOR
B25E1092

Q Write a C++ program to print pascal triangles.

```
#include <iostream>
using namespace std;

int main() {
    int rows;
    cout << "Enter the number of rows: ";
    cin >> rows;

    for (int i = 0; i < rows; i++) {

        for (int space = 1; space <= rows - i; space++) {
            cout << " ";
        }

        int val = 1;
        for (int j = 0; j <= i; j++) {
            cout << val << " ";
            val = val * (i - j) / (j + 1);
        }
        cout << "\n";
    }
    return 0;
}
```

OUTPUT :

Enter the number of rows: 3

```

    1
  1  1
1  2  1
```

=== Code Execution Successful ===

Q Find out the sum of even numbers 1 to 50 using loop

```
#include <iostream>
using namespace std;

int main() {
    int i, n=50, sum=0 ;
    cout<< "Even numbers are: "" \n";
    for (int i=2; i<=n; i++){
        cout<<" "<< i ;
    }
    for (int i = 2; i <= n; i += 2)
    {
        sum += i;
    }

    cout << "\n""The sum of even numbers from 1 to " << n << " is: " << sum ;
    return 0;
}
```

OUTPUT:

```
Even numbers are:
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27
 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49
 50
The sum of even numbers from 1 to 50 is: 650
```

Q. Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

```
#include <iostream>
using namespace std;

// Function declarations
void inputArray(int arr[], int size);
void displayArray(int arr[], int size);
void sortArray(int arr[], int size);
void swapNumbers(int &a, int &b); // Added swap function declaration

int main() {
    int arr[2], i;
    int size = sizeof(arr) / sizeof(arr[0]);

    cout << "\nNumber of elements in array: " << size << endl;

    inputArray(arr, size);

    cout << "\nThe elements in the array are: ";
    displayArray(arr, size);

    // First sort the array
    sortArray(arr, size);
    cout << "\nThe elements in the array after sorting are: ";
    displayArray(arr, size);

    if (size == 2) {
        swapNumbers(arr[0], arr[1]);
    }
    cout << "\nThe elements in the array after swap are: ";
    displayArray(arr, size);

    return 0;
}

void inputArray(int arr[], int size) {
    cout << "\nEnter " << size << " integers: " << endl;
    for (int i = 0; i < size; i++) {
        cin >> arr[i];
    }
}
```

```

    }
}

void displayArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

void sortArray(int arr[], int size) {
    if (size == 2 && arr[0] > arr[1]) {
        swapNumbers(arr[0], arr[1]);
    }
}

void swapNumbers(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

```

Number of elements in array: 2

Enter 2 integers:

2 8

The elements in the array are: 2 8

The elements in the array after sorting are: 2 8

The elements in the array after swap are: 8 2

=== Code Execution Successful ===

Q WRITE A C++ PROGRAM TO EXCEPT THE ARRAY OF N NUMBERS FROM THE USER AND DISPLAY THE SAME ARRAY TO THE USERS

```
#include <iostream>
using namespace std;

int main()
{
    int i, arr[7];

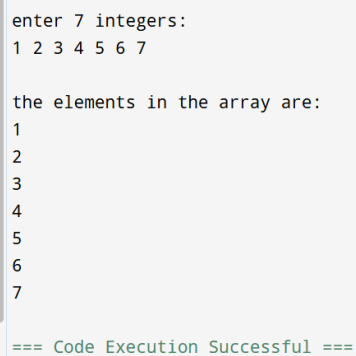
    cout << "\nenter 7 integers:" << endl;

    for(i = 0; i < 7; i++)
    {
        cin >> arr[i];
    }

    cout << "\nthe elements in the array are:";

    for(i = 0; i < 7; i++)
    {
        cout << "\n" << arr[i];
    }

    return 0;
}
```



The screenshot shows the output of the C++ program. It starts with the prompt "enter 7 integers:" followed by the user input "1 2 3 4 5 6 7". Then, it displays "the elements in the array are:" followed by the numbers 1 through 7, each on a new line. At the bottom, it shows the message "=== Code Execution Successful ===".

```
enter 7 integers:
1 2 3 4 5 6 7

the elements in the array are:
1
2
3
4
5
6
7

=== Code Execution Successful ===
```